

EE5311 Digital IC Design – Assignment 4

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Summary of Results

Item	Configuration	Delay / Period	Frequency
Q1(a)	2× inverter, ideal	19.95 ps	–
Q1(a)	2× inverter, extracted (per-cell)	25.26 ps	–
Q1(b)	inv2 layout, extracted (net V_{out} incl.)	26.30 ps	–
Q2(a)	7-stage ring osc., extracted, $V_{DD} = 1.8V$	558.1 ps	1.792 GHz
Q2(c)	7-stage ring osc., ideal, $V_{DD} = 1.8V$	409.1 ps	2.444 GHz

1 Inverter Layout – DRC & LVS

Reference inverter ($W_p/L_p = 0.84/0.15\mu m$, $W_n/L_n = 0.42/0.15\mu m$) laid out in Sky130; verified DRC/LVS clean before use in the delay and oscillator extractions below.

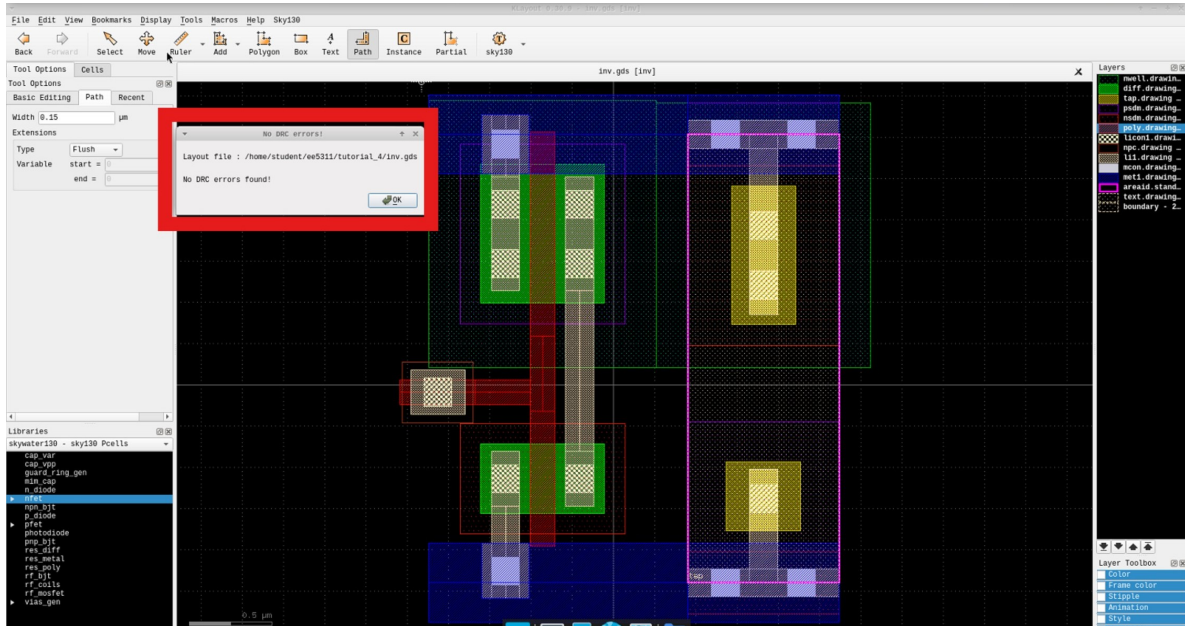


Figure 1: Inverter layout – DRC clean.

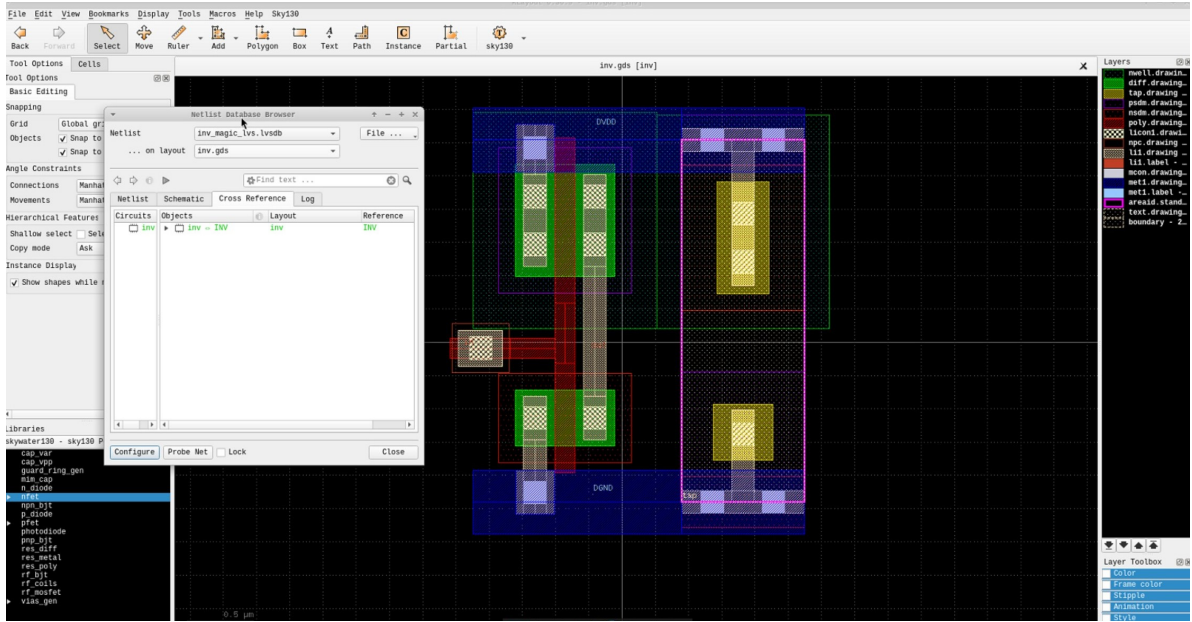


Figure 2: Inverter layout – LVS clean (inv $\leftrightarrow$ INV match).

Check	File	Result
DRC	inv.gds	No errors found
LVS	inv.gds	Match (inv $\leftrightarrow$ INV)

2 Q1(a): Delay – Ideal vs. Extracted (per-cell)

Two cascaded inverter instances (x1, x2) simulated first with ideal device characteristics, then with each instance individually replaced by its own extracted (parasitic) netlist. Propagation delay = $(t_{HL} + t_{LH})/2$, measured at the 0.9 V crossing.

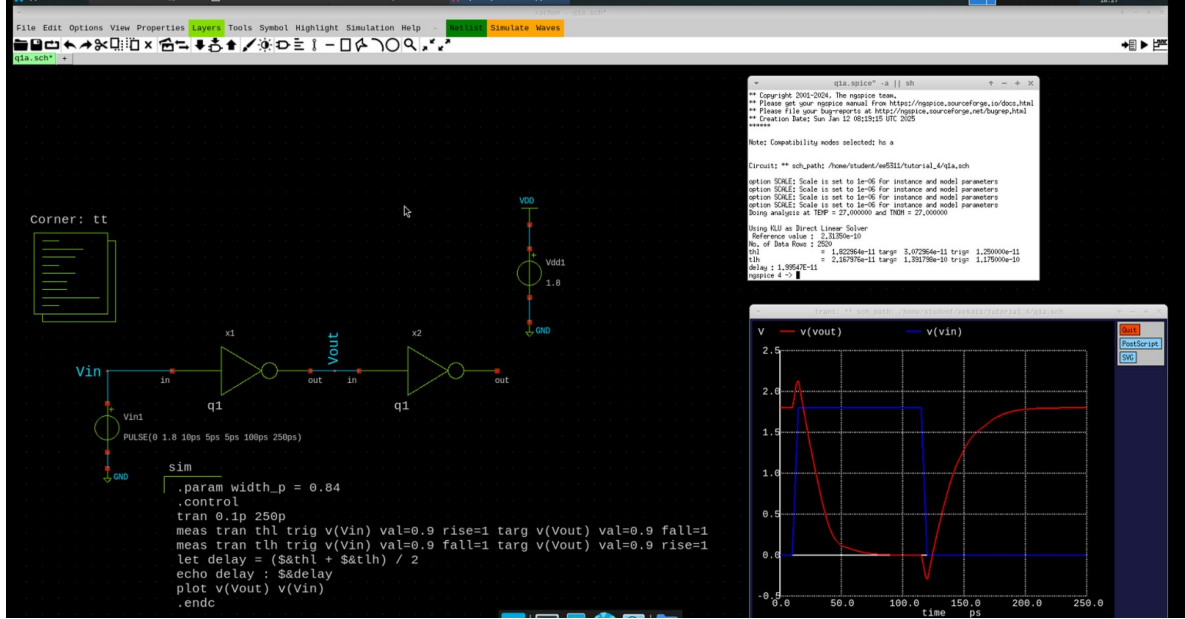


Figure 3: Ideal characteristics: delay = 19.95 ps.

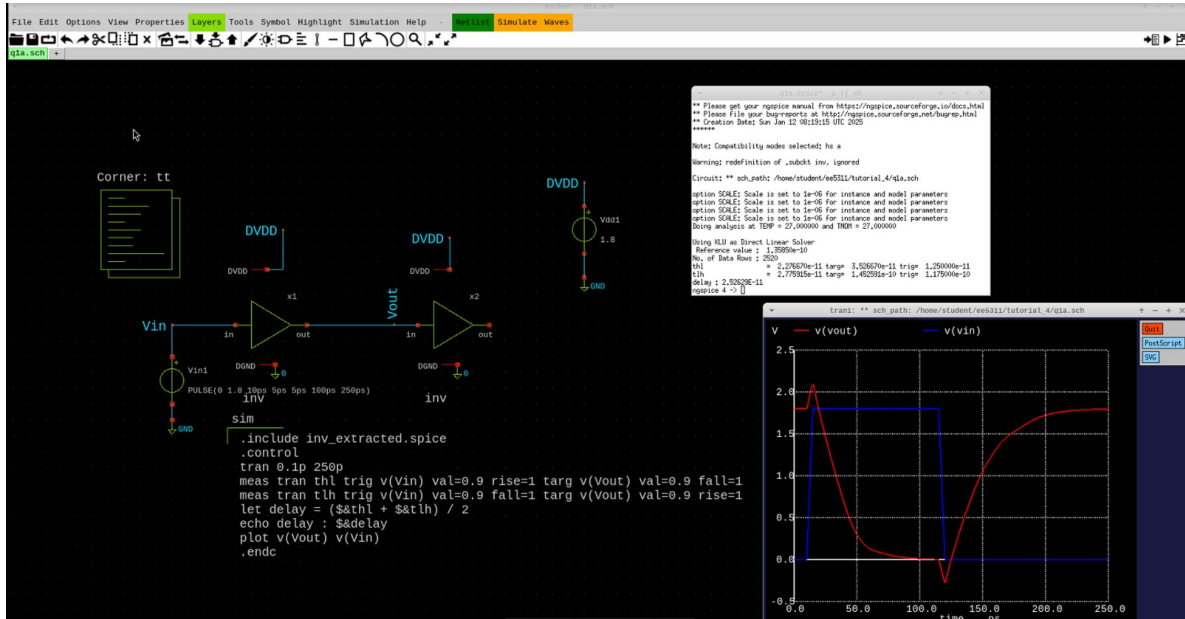


Figure 4: Extracted netlist (both instances): delay = 25.26 ps.

Case	t_{HL}	t_{LH}	Delay
Ideal	18.23 ps	21.68 ps	19.95 ps
Extracted (per-cell)	22.77 ps	27.76 ps	25.26 ps

3 Q1(b): Delay Including Layout Parasitics of Net V_{out} (inv2)

Both inverters laid out as a single `inv2` block so the extracted netlist also captures the interconnect (net V_{out}) parasitic, not just the two devices individually; delay measured the same way as Q1(a).

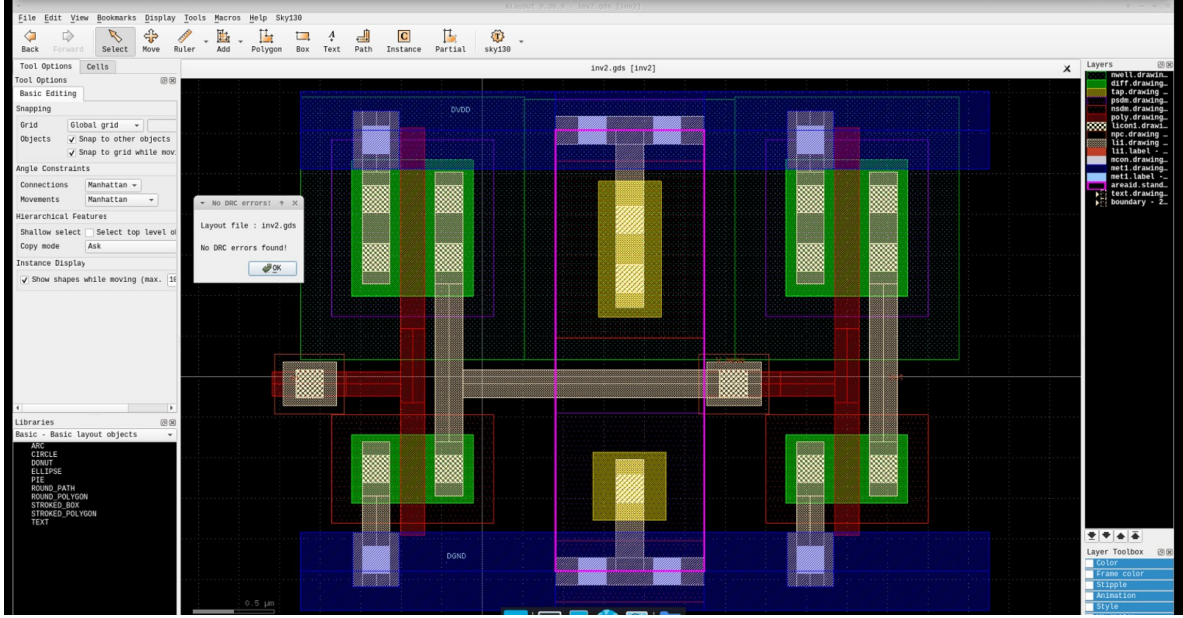


Figure 5: inv2 layout – DRC clean.

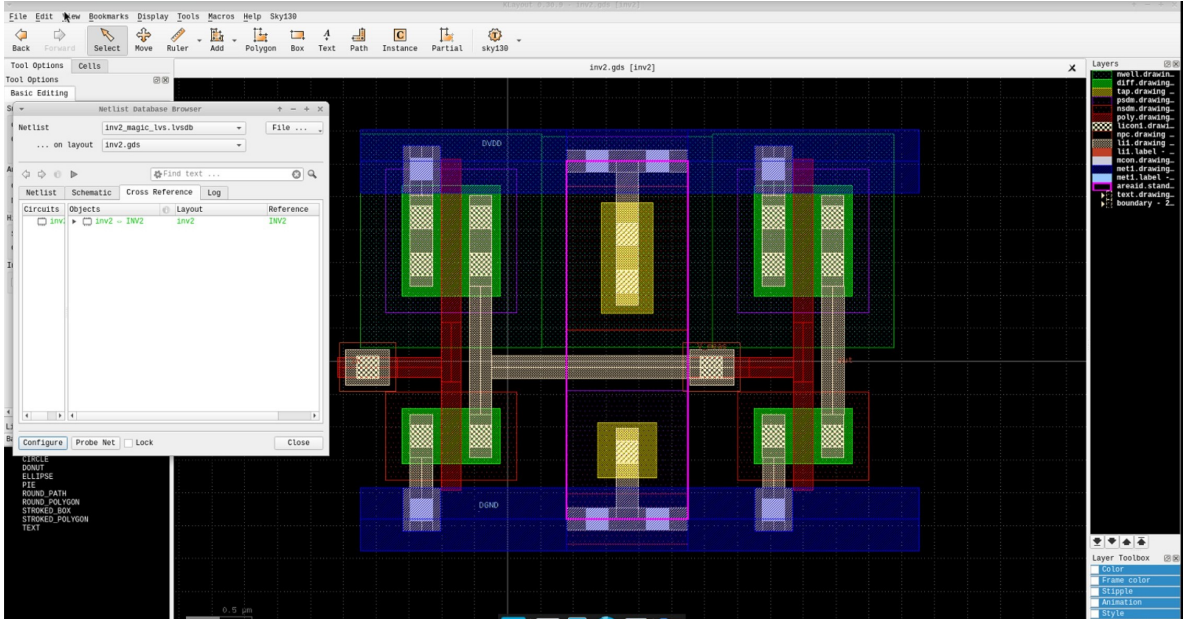


Figure 6: inv2 layout – LVS clean (inv2 ↔ INV2 match).

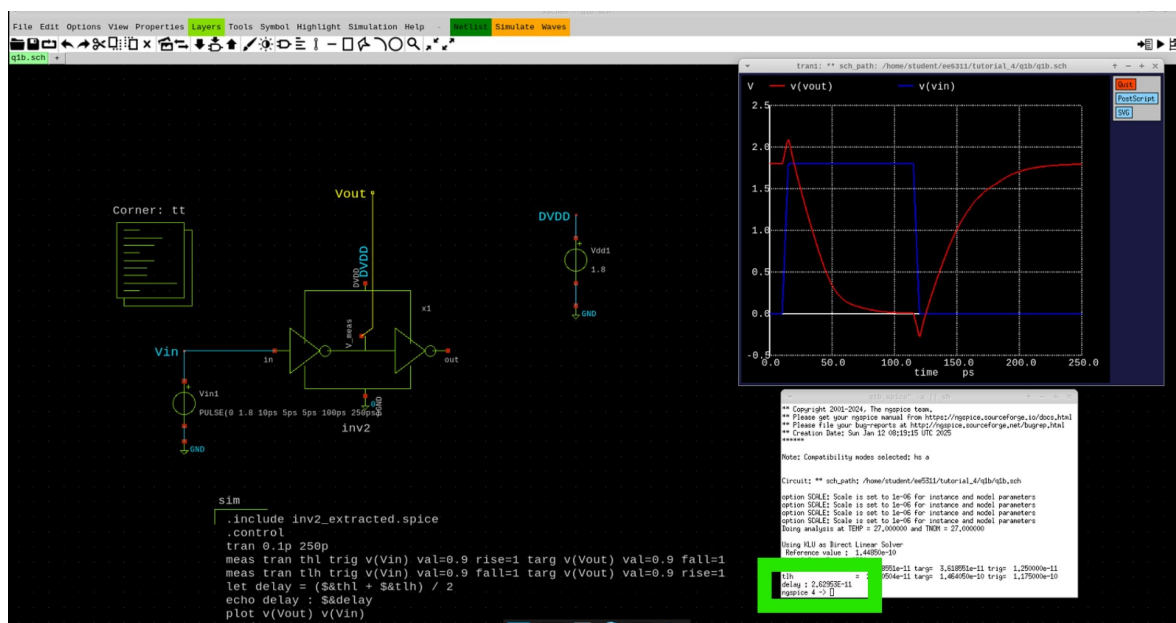


Figure 7: Extracted netlist (single inv2 layout): delay = 26.30 ps.

Check	File	Result
DRC	inv2.gds	No errors found
LVS	inv2.gds	Match (inv2 $\leftrightarrow$ INV2)

t_{HL}	t_{LH}	Delay
23.69 ps	28.91 ps	26.30 ps

Net-parasitic-inclusive delay (26.30 ps) is slightly higher than the per-cell extracted delay (25.26 ps) since the interconnect capacitance/resistance of V_{out} is now also accounted for.

4 Ring Oscillator Layout – DRC & LVS

7-stage ring oscillator built from the inverter cell above; layout verified DRC/LVS clean before parasitic extraction.

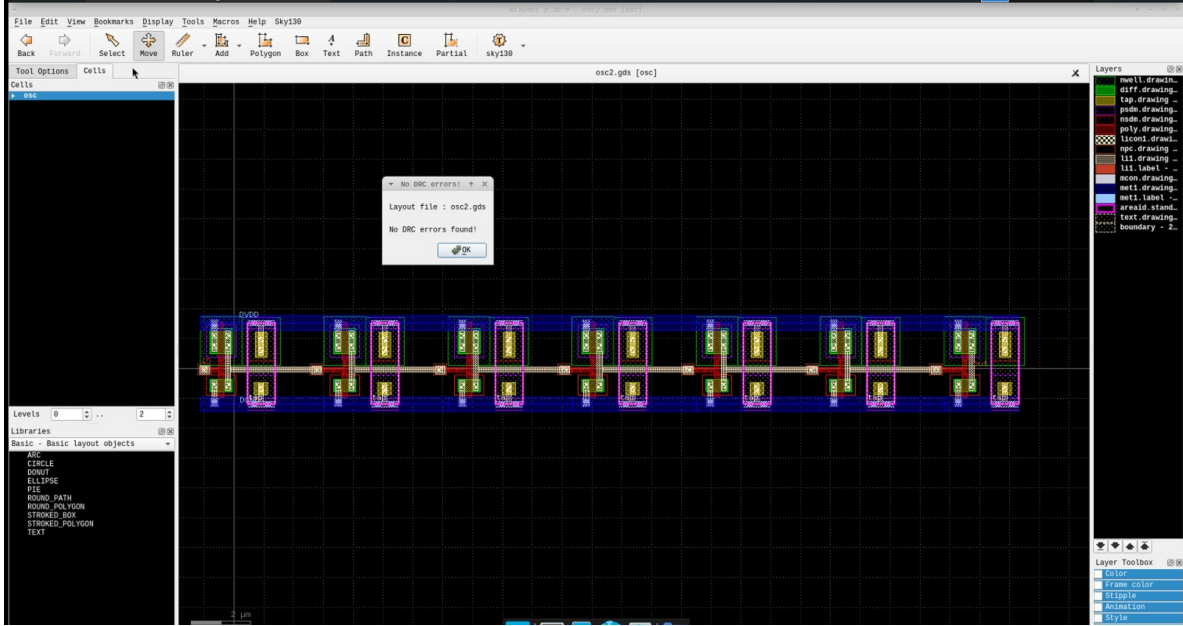


Figure 8: 7-stage ring oscillator layout – DRC clean.

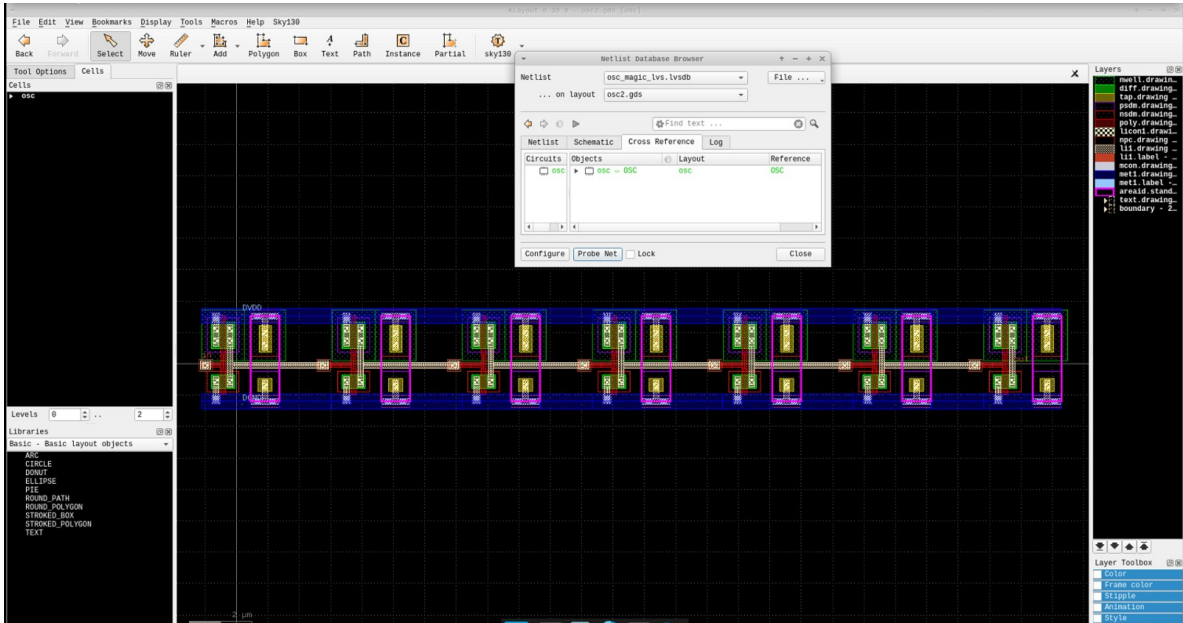


Figure 9: 7-stage ring oscillator layout – LVS clean (osc ↔ OSC match).

Check	File	Result
DRC	osc2.gds	No errors found
LVS	osc2.gds	Match (osc $\leftrightarrow$ OSC)

5 Q2(a)/(b): Frequency & Period vs. V_{DD} – Extracted

Ring oscillator simulated with the extracted (layout-parasitic) netlist for $V_{DD} = 1.0$ to 1.8 V in 0.1 V steps (V_{out} initialized to 0 V); period/frequency computed from the 3rd and 4th rising crossings of V_{out} at 0.4 V.

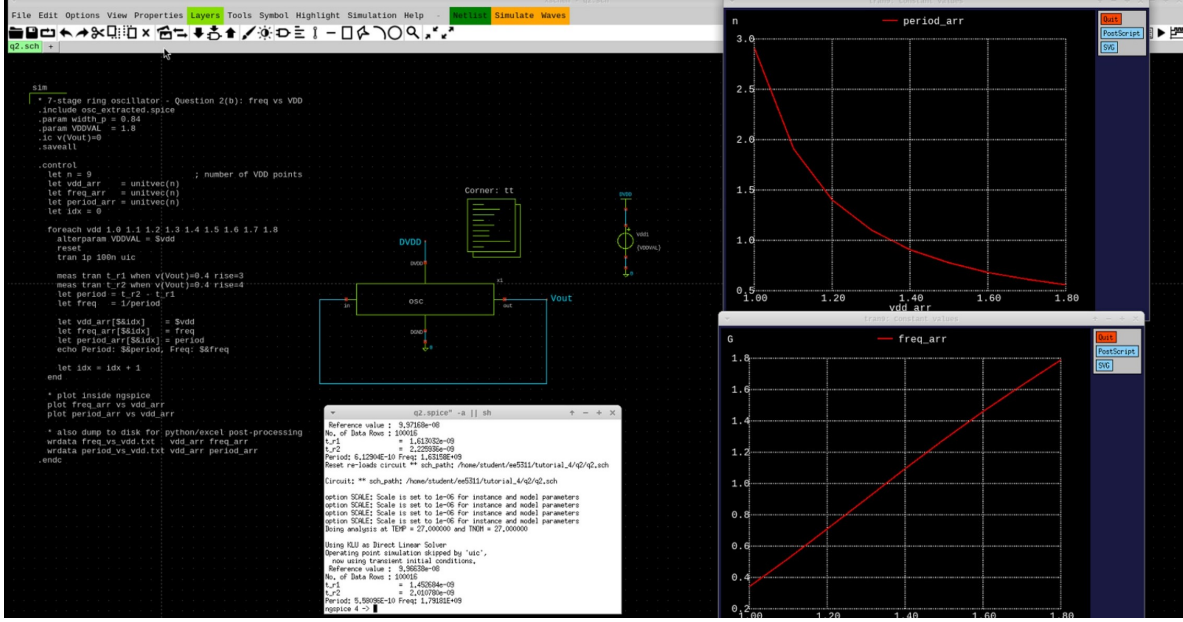


Figure 10: Post-layout extracted: period_arr and freq_arr vs. V_{DD} (1.0–1.8 V).

V_{DD}	Period	Frequency
1.7 V	612.90 ps	1.632 GHz
1.8 V	558.10 ps	1.792 GHz

(Last two swept points read directly from the console; full 1.0–1.8 V sweep is plotted above.)

6 Q2(c): Comparison with Pre-Layout (Ideal) Simulation

Same sweep repeated with ideal (non-extracted) device characteristics. Layout parasitics reduce the oscillation frequency by $\sim 27\%$ at $V_{DD} = 1.8\text{V}$ (1.792 GHz extracted vs. 2.444 GHz ideal).

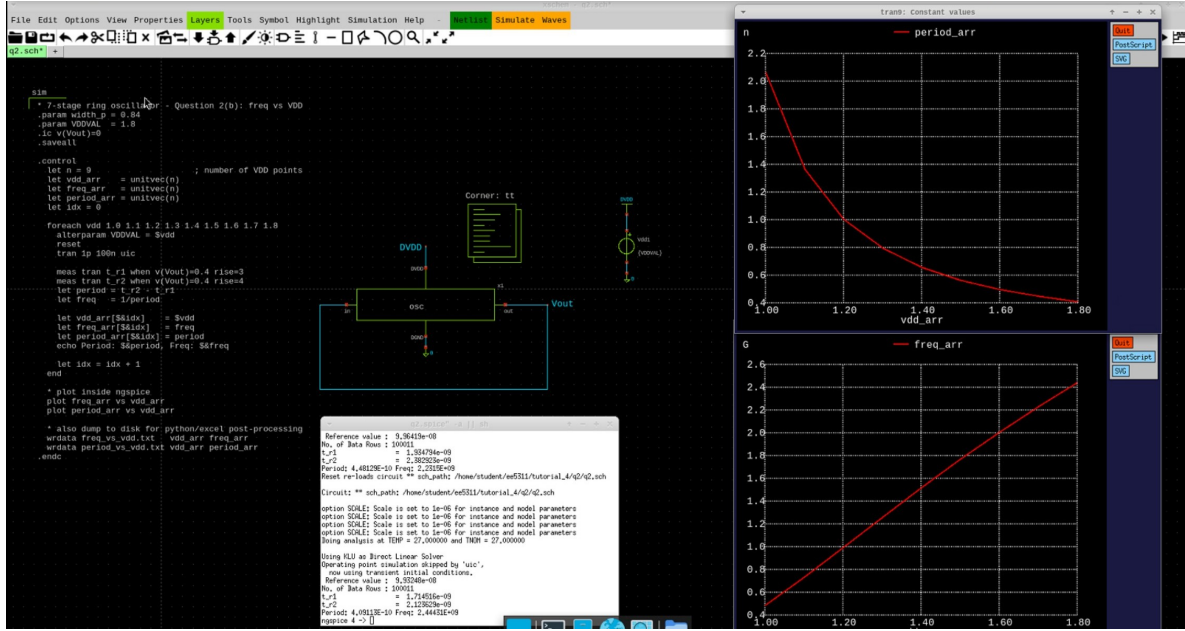


Figure 11: Ideal (pre-layout): period_arr and freq_arr vs. V_{DD} (1.0–1.8 V).

V_{DD}	Period	Frequency
1.7 V	448.13 ps	2.232 GHz
1.8 V	409.11 ps	2.444 GHz

(Last two swept points read directly from the console; full 1.0–1.8 V sweep is plotted above.)